

required for conformal placement of devices on the human body or any arbitrary surface.

Technological developments and challenges

The interest in fabrication strategies for stretchable electronics has been driven by the huge demand for wearable, intelligent and integrated electronic systems in the past decade. Therefore strenuous efforts have been made to realise the stretchability of electronic devices and reduce the total fabrication cost via sophisticated material design and novel device configuration.

However, mechanical compliance is critical in the manufacture of stretchable electronic devices, as devices should not incur physical damage or change their performance under bending or stretching states. A number of important discoveries and breakthroughs in the fundamentals of this technology and fabrication strategies are taking place to solve these challenges and achieve the above-mentioned targets.

Various strategies are used to make electronic devices stretchable, including directly utilising intrinsically stretchable materials or constructing structure design strategies such as wavy structure configuration, island-interconnect configuration, mesh structure, fractal design approach, origami and kirigami structural configurations.

However, stretchable conductors and electrodes are the fundamental building blocks for final stretchable electronic devices. The compliant, deformable and stretchable qualities of stretchable conductors have updated traditional ideas that people must use non-flexible silicon-based electronics. This has opened up a new direction of next-generation flexible electronics.

Development of stretchable electronic devices is not a new concept, and a wide range of stretchable electronic devices are being investigated. Further growth in such devices could have a wide range of potential uses, from wearable health monitors and elastic solar cells to artificial skin. But there are several challenges involved in finding suitable materials and manufacturing methods. The biggest challenge for making stretchable electronics is that each component must



Wearable electronics (Credit: <http://nwctrail.com>)

endure being compressed, twisted and applied to uneven surfaces while maintaining its performance.

Primary developing trends in flexible and stretchable electronics involve the innovation of material synthesis, mechanical design and fabrication strategies that employ soft substrates. **The biggest challenge here is that the entire electronics system must allow not only bending but also stretching. Therefore stretchable conductors become a crucial construction unit for the connection of working circuits of various stretchable devices.**

Owing to the success of stretchable conductors, various stretchable electronic devices are fabricated with the help of multiple manufacturing strategies, including stretchable heaters, stretchable energy conversion and storage devices, stretchable transistors, sensors and artificial skin. Continuous development of stretchable electronics has led to the new functionality of transparency, and fabrication of transparent, stretchable electronic devices has gained a lot of interest due to the potential of wearable electronic systems.

For instance, low-cost stretchable conductors and electrodes are being made from silver nanowires and graphene. Presently, an electronic thin-film transistor matrix can be stretched to about 140 per cent, and its transistor parameters also remain unchanged upon stretching.

An urgent technical problem is the need for stretchable energy conversion

and storage devices such as batteries. Zinc-based batteries are promising candidates; however, more work is required to make them commercially viable. An alternative to batteries is stretchable nanogenerators, which can produce electricity from various freely available vibrations, such as wind or human body movements. Stretchable solar cells could also be used to power wearable electronic devices.

By integrating multiple stretchable components—such as temperature, pressure and electrochemical sensors—it is possible to create a material resembling human skin that could use signals from sweat, tears or saliva for real-time, non-invasive healthcare monitoring, as well as for smart prosthetics or robots with enhanced sensing capabilities. However, at present, fabrication of artificial skin remains time-consuming and complex.

Currently, there are two main strategies for manufacturing stretchable electronics. The first is to use intrinsically stretchable materials, such as rubber, which can endure large deformations. However, these materials have limitations such as high electrical resistance.

The second method is to make non-flexible materials stretchable using innovative design. For example, brittle semiconductor materials like silicon can be grown on a pre-stretched surface and then allowed to compress, creating buckling waves.

In the future, stretchable electronics may be enhanced with new capabilities